1. ~~Fa~~ A heap is a complete binary tree, so a heap of height h would have at least 2^{h-1} ~~nod~~ elements and ~~one~~ at most $2^h - 1$ elements

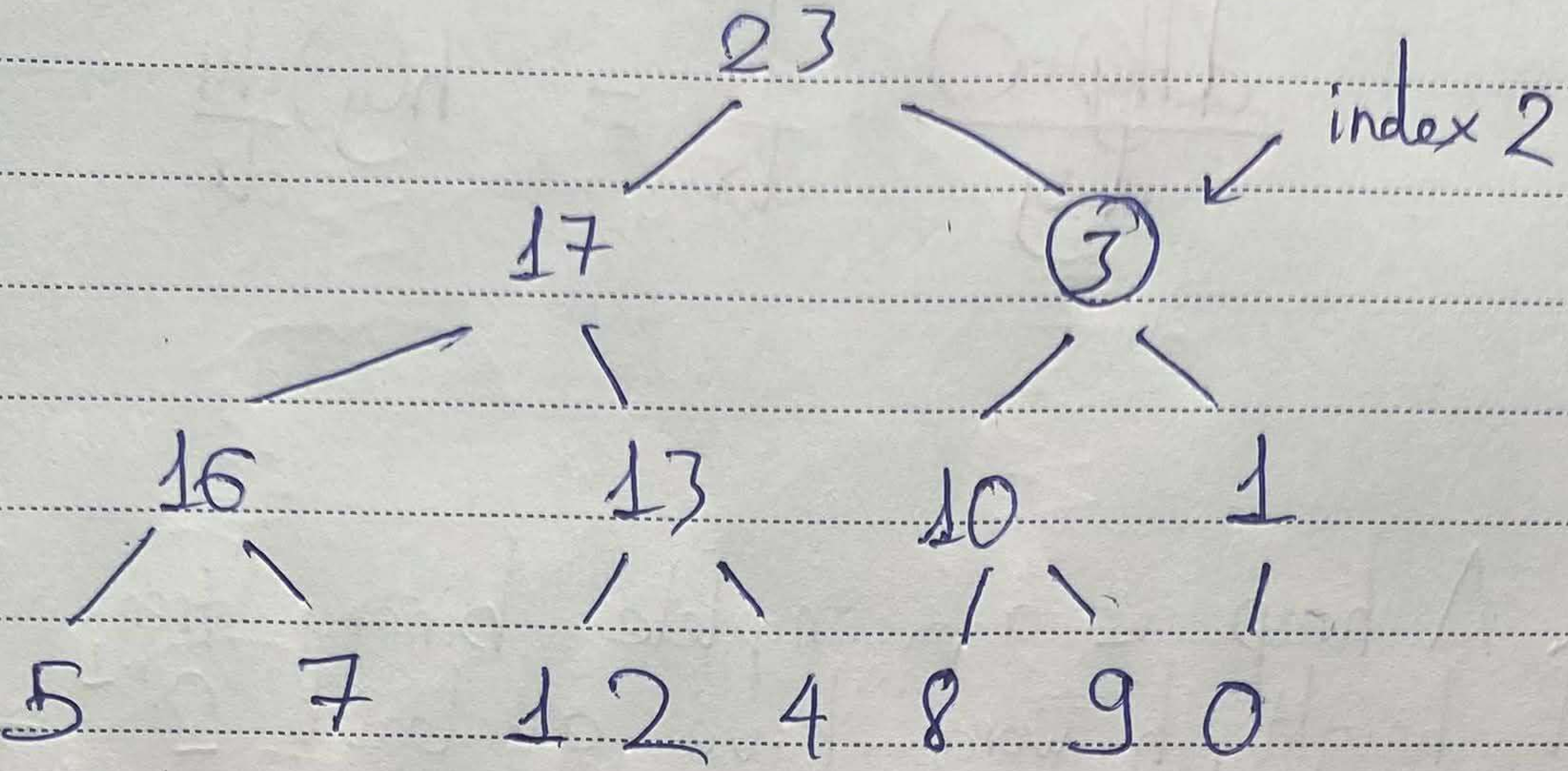
2. The smallest element in a max heap may reside in any of the "leaf nodes" elements

3. Every path from the top element to a "leaf node" element secures the ascending order ~~to~~ so it is also a min heap.

4. No, as the order suggests, 7 is a child element of 6 while 7 > 6 so this is not a max-heap.

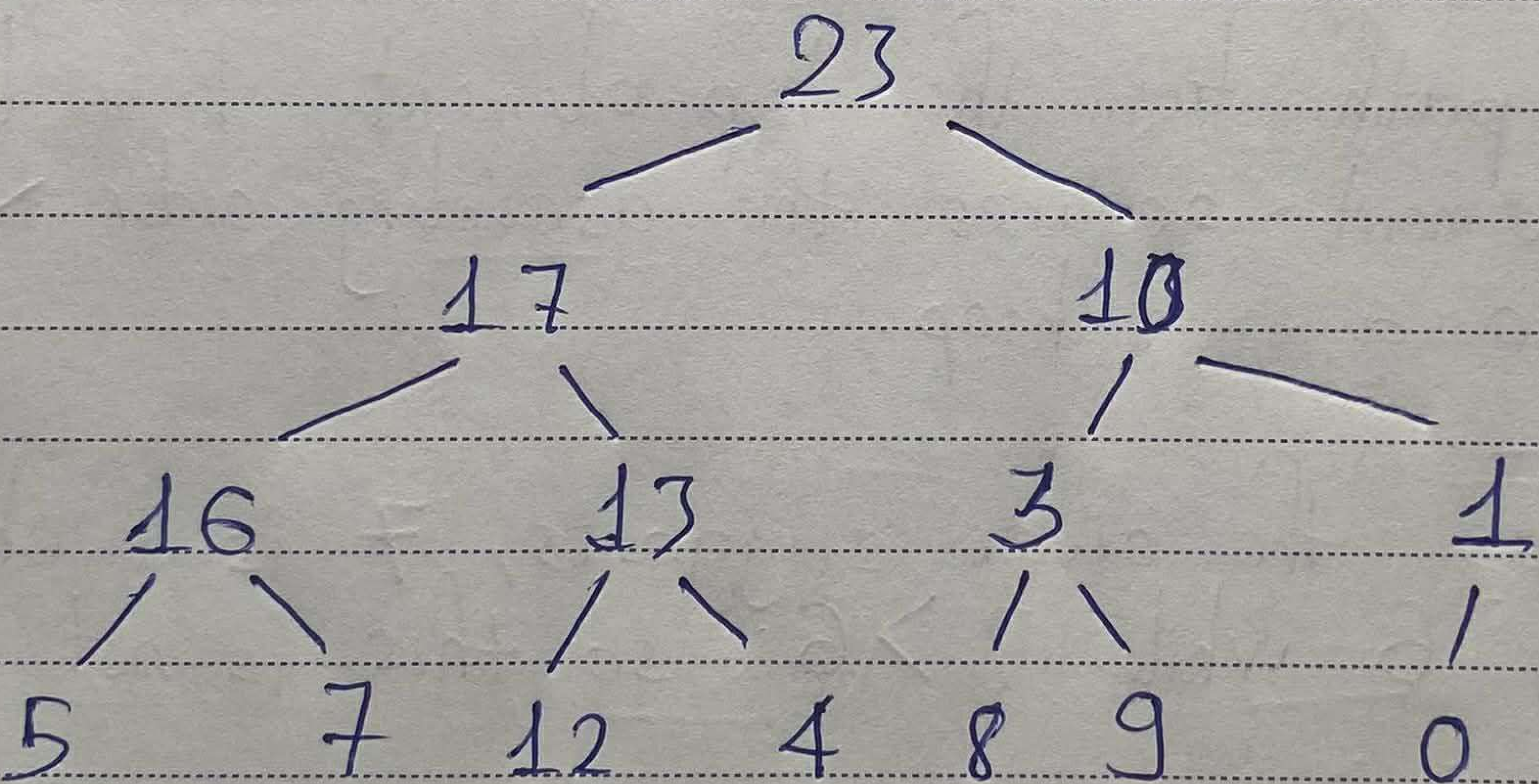
6. MAX-HEAPIFY, as the name suggests, will do the heapify operation following the max-heap rules, which is swapping an element with one of its children to maintain the descending order. Now if the current element is larger than both of its children, the order is not broken and the command does nothing.

5.



We have $\begin{cases} 3 < 10 \\ 10 > 1 \end{cases}$

So we swap 3 with 10 to maintain the descending order:



Follow the similar logic, we swap 3 with 9.